Camden Council

DESIGN AND CONSTRUCTION SPECIFICATION

ACCESS DRIVEWAYS

(ASSOCIATED WITH AN APPLICATION FOR A NEW BUILDING)

Camden Council

Specification for Access Driveways

General

Aim of this Specification

Planning Design and Construction

The specification for Access Driveways aims to provide some simple guidelines, processes and requirements in the planning and construction of vehicle access from the public road to the private property.

Separate Approval required?

This approval does not approve driveways which are not associated with an application for a new building, second driveways or driveways in a heritage area. A separate approval is required for additional driveways and driveways connected to or within a heritage curtilage.

Procedures

The specification details the design, approval, inspection and construction procedure.

Description of Work

An access driveway is a constructed vehicular access across public land from the kerb and gutter or road shoulder to the property boundary.

Types of Access Driveways

There are three types of access driveways:

a) Urban Residential access driveway (in residential areas only)
b) Rural Residential access driveway (in rural areas only where kerb and gutter does not exist)
c) Industrial and Commercial access driveways to comply generally with this specification and subject to engineering design which is to be prepared by the applicant and submitted with the development application for the project.

Second Driveways

If not approved as part of a Development Application all additional driveways require a separate approval, this approval must be obtained prior to works commencing.

Owner's Responsibility

The construction of an access driveway from the road to the property boundary is the responsibility of the land owner/builder including construction costs and inspection fees, however applications and fees may be paid by building or concrete contractors.

Public Safety

Council governs the specification and inspection of all access driveways to ensure public safety and construction standards are observed within the public footpath area.

Australian Standards (AS)

Australian Standards

- AS 1742.3: - "Manual of Uniform Traffic Control Devices – Traffic control devices for works on roads".
- AS 1742.10: - "Manual of Uniform Traffic Control Devices – Pedestrian control &

protection".
- AS 4586: – "Slip resistance classification of new pedestrian surface materials"
- AS 3661.1: – "Slip resistance of pedestrian surfaces – Requirements"
- AS 3661.2: – "Slip resistance of pedestrian surfaces – Guide to the reduction of slip hazards"
- AS 2890.1: – "Parking Facilities – Off-street car parking"

NSW Acts

NSW State Legislation

- Roads Act 1993

Applications, Fees and Inspections

All work constructed in the public road reserve must be undertaken by a competent and approved contractor with relevant experience in this type of construction. The contractor is required to be licensed with the Department of Fair Trading.

Competent and Approved Contractor

Council may direct work in the road reserve to be removed and/or reinstated if work is carried out without approval or supervision.

Unsatisfactory and Unauthorised Works

Council officers can be contacted at any time to give particular guidance on all the issues below.

Advice

A list of useful contact details can be found at the end of this document.

Inspections

Preliminary Inspection

In order to ensure an acceptable standard of construction, an inspection is required after excavation and preparation of the access driveway and prior to placement of the pavement material. This usually involves inspecting formwork, reinforcement, joining material, bedding sand along with a check of levels, width, thickness and gradients prior to pouring of concrete to ensure compliance with Council's standards. Upon completion of works across the footpath, a final inspection is required to ensure the site has been restored.

Final Inspection

Inspections will only be carried out during Council's normal working hours and 24 hours notice is required.

Inspection Notice

Preliminary and Design Considerations

Public Service Utilities

Public Service utilities such as telephone, electricity, water, sewer and gas are often located in the footpath area. Access to the site should not interfere with the existing public utility infrastructure, including Council drainage structures, unless prior approval is obtained from the relevant authority. Such approval may require the relocation of the service fixture or provision of trafficable covers at no cost to Council.

Ensure No Interference With Public Services

Driveways are to be a minimum of 1.0 metre clear of all public road drainage structures and 0.3m clear of any kerb roofwater outlet.

Ensure no Interference With

Driveways are to be located a minimum of 0.3m clear of any pram ramp.

Infrastructure

You should advise Integral Energy, Sydney Water, all telecommunication providers who service the area (e.g Telstra, Optus), and AGL prior to commencement of works to check whether they have any specific requirements to facilitate future provision of services. This may ensure your driveway does not need to be excavated in the future.

Check Service Authority Requirement

Integral Energy require a minimum setback of 1.0 metre from the edge of the access driveway to any power pole and 450mm from an above ground service pillar. This is to enable maintenance and to reduce the risk of vehicle impact.

Power Pole & Pillar Setback

The location of all services, telephone, electricity, water, sewer and gas should be verified with the relevant authorities before work commences. Interfering with them can be dangerous and costly. Any alteration or damage incurred to these services is the contractor's responsibility. A check of underground service location can be done by calling "Dial Before You Dig" on 1100.

Check Service Locations Prior to Excavation

Design and Safety Considerations

90° Intersection With Road

Access should be safe, direct and available at all times. The access should intersect the road at 90°. Council consultation will be required where this is not achievable and details shall be submitted with the Application.

On corner lots, the closest edge of the access point should be setback a minimum of 6.0 metres from the tangent point to a road intersection.

Corner Lot Setback

On busy roads, roads with poor sight distance or on steep driveways, entry and exit should always be in a forward direction. This will require the construction of a manoeuvring bay adjacent to the parking area. The manoeuvring bay shall ensure all turning movements are done entirely within the private property.

Access in a Forward Direction

Vegetation trimming and embankment benching adjacent to the access point may be required to provide adequate and safe sight distance. Note that Tree Preservation Orders may apply to some vegetation and Council liaison is required prior to the commencement of work.

Sight Distance

When excavation across the footpath is necessary, the batters on either side of the driveway shall not be steeper than 1 (vertical):6 (horizontal).

Maximum Batter 1:6

Where a concrete footpath does not exist fronting the property, it is important to ensure the surface gradient of the footpath allocation is 4.0%. This allows future footpath construction adjacent to the driveway without having an excessive crossfall. Note that the footpath allocation area starts 900mm (800mm within Oran Park) from the property boundary and is 1200mm wide. A pre-commencement inspection must be arranged with Camden Council if the above 4% can not be achieved.

4% Footpath Crossfall

Where an existing concrete footpath is affected by an access driveway, the relevant section is to be removed and reconstructed. Council **must be consulted prior to any works commencing**. All costs for the works associated with footpath alterations shall be at no cost to Council.

Existing Concrete Footpath

Council approval is required for the visual appearance of the driveway. For all new dwellings, details are to be submitted with the development application. The following finishes are permissible subject to full compliance with all other parts of this

Finished Visual Appearance

specification:

- Plain concrete
- Exposed aggregate
- Coloured concrete
- Stamped concrete

No colour in layback

All laybacks shall be constructed from plain uncoloured concrete

Future Restoration

Note: Any future replacement of the access driveway undertaken by Council or any other public authority will only be restored in plain concrete. The additional cost to restore to any other featured colour or surface finish shall be at no cost to Council.

Non-Slip Finished Texture

Access driveways must have a non-slip finish and provide traction for pedestrians and vehicles. The surface must comply with the requirements of the following Australian Standard (AS)

- AS 4586: – "Slip resistance classification of new pedestrian surface materials",
- AS 3661.1: – "Slip resistance of pedestrian surfaces – Requirements"
- AS 3661.2: – "Slip resistance of pedestrian surfaces – Guide to the reduction of slip hazards".

Single Access Point

Driveway access to the street should generally be confined to a single point in order to maintain street parking, landscaping opportunities on the street and within the building setback and minimise impact upon the streetscape. A separate approval is required for a second driveway, please contact council prior to works commencing.

Design Plan Information

All New Developments require Access Driveway Details With The DA

For all new developments – residential, industrial or commercial – plans and details of the proposed access driveway are to be submitted with the Development Application (DA). If a DA is not required (eg, construction of a driveway to an existing dwelling), the access driveway and driveway requirements should be constructed in accordance with this specification.

On flat sites a plan view showing any existing features such as street drainage pits, on-street parking bays, street traffic control devices (e.g roundabouts, thresholds, planter boxes), footpaths service poles, signage, trees etc. with a typical cross section and brief specification statement is generally adequate.

Difficult Sites Require More Detailed Information and a Site Specific Design

On steeper sites the designer should provide the following additional information with the Development Application:

- Contour plan with spot levels.
- Plan view of the proposed development showing the access alignment relating to existing and identifiable features.
- Longitudinal section of the driveway from the centreline of the road to the rear of the parking platform. Transitions and grades are to be detailed in accordance with AS2890.1.
- A typical cross section and cross sections at logical intervals.
- Pavement detail. This should include pavement type, thickness, width, surface finish and sub-structure. Expansion joint, control joint and key joint detail should be included if the proposed driveway is a concrete construction.
- Proposed method of drainage of the driveway surface.
- Extent of area to be disturbed.

- Extent of cut and fill including retaining wall details.
- Hand rail, safety fence & wheel stop detail.
- Vehicle turning path detail.
- Location of and impact on public utilities.
- Impact on natural features.
- Pedestrian access from the parking space to the dwelling.
- Proposed restoration details.

General

Ensure Public Safety is Upheld

All crossings shall be barricaded and lit with lamps to ensure they are safe and to prevent them from being used for four days after completion. Care should be taken to ensure that open excavations are barricaded and lit with lamps and that safe alternate pedestrian access around the excavation is provided at all times. Australian Standards 1742.3:1996 - "Manual of Uniform Traffic Control Devices – Traffic control devices for works on roads" and 1742.10:1990 - "Manual of Uniform Traffic Control Devices – Pedestrian control & protection" shall be observed at all times whilst working within public land.

Australian Standards

Stormwater Protection

Precautions must be taken during construction to prevent the site being inundated from roadwater.

Provision of Site Stormwater Drainage

Where a gutter layback and/or access driveway interferes with a stormwater pipe and/or its outlet through the kerb, the stormwater pipe must be carried diagonally across the footpath to connect with a new kerb outlet which is to be provided by the contractor. All drainage lines under access driveways and drainage weepholes through the face of kerb must be 100mm x 50mm galvanised rolled hollow section (RHS) at minimum grade of 0.5%. The invert of the drainage hole is to be 10mm above the gutter invert.

Damage to Council's Assets

All access construction works and public utility relocation shall incur no cost to Council and is to include any necessary work to make the construction effective. The contractor and/or the landowner is liable for any damage to Council's assets. The contractor and/or the landowner shall arrange with Council to make good the damage, at no cost to Council, prior to Council issuing a final clearance for the works.

No Footpath Doweling

All laybacks are to be a separate construction to access driveways. Laybacks should not be dowelled into the footpath crossing

Existing kerb and gutter

Where vertical kerb and gutter (Barrier Kerb) exists at the proposed access point, the kerb and gutter must be totally removed and replaced with an integral Gutter Crossing. No bitumen is to be removed.

Redundant Layback

Where a redundant layback will occur at the frontage of the property, new concrete kerb & gutter to Council's specification and approval is required to replace the redundant layback. Such kerb and gutter shall match the adjacent type in the immediate vicinity.

Ensure No Trip Edges

Any disturbed areas are to be back-filled, compacted and restored to ensure a smooth transition with no trip edges. Stockpiling of any material is not permitted on the roadway or nature strip.

Erosion and Sedimentation Control

All disturbed areas will need to be effectively controlled during construction in terms of sedimentation/erosion controls and then turfed upon the completion of works. Council's Sedimentation and Erosion Control Code of Practice and Policy should be followed.

Surface Gradient

The top surface of the access driveway shall rise in a straight line on a grade of 4% from the back of the kerb crossing to the property boundary. The slab shall be a constant width to the boundary and at right angles to the kerb and gutter where ever possible.

4% Footpath Crossfall

You must contact Council to arrange a pre-commencement inspection if a concrete footpath exists across the proposed driveway or if the above conditions are impractical. It is important that the grades above are adhered to, to facilitate future footpath construction.

If Concrete Footpath Exists, Contact Council

Urban Residential Access Driveways

Access driveways and driveways shall have a minimum width of 4.0m and maximum width 5.5m.

Width

Concrete Strength, Thickness Reinforcement Subgrade, Expansion Joints

Concrete Driveways

- Concrete strength to be 32 MPa minimum
- Slab thickness to be 125mm
- Reinforcement to be SL72 fabric
- Concrete to be placed on suitably compacted sub-grade and 50mm of bedding sand.
- Expansion joints are required to isolate the slab from other concrete work. Expansion joints shall be placed at the kerb crossing and property boundary. Joints shall be made from 10mm thick bitumen impregnated fibreboard over the full joint area and depth of the slab. A sawcut is also an acceptable expansion joint at the property boundary.
- All edges shall be finished round with a coarse surface finish to provide traction.

Round Edges

Rural Residential Access Driveways

Rural access driveways are required in rural areas where kerb and gutter does not exist. The access driveway shall be constructed in accordance with the specification for Rural Residential Driveways.

Rural Areas

Rural Dish Crossing

A Council standard concrete dish crossing is required to be constructed. The alignment of the dish crossing is governed by existing kerb & gutter and other dish crossings in the adjacent area. If none exist, the invert of the dish crossing should be aligned with the invert of the table drain in the road shoulder.

Location and Alignment of Dish Crossing

Rural Pipe Crossing

Rural Pipe Crossing

Pipe Crossings may only be permitted in rural areas where a dish crossing would not provide a suitable access to the allotment. Installation of a pipe crossing is subject to Council approval.

Bitumen Shoulder Seal

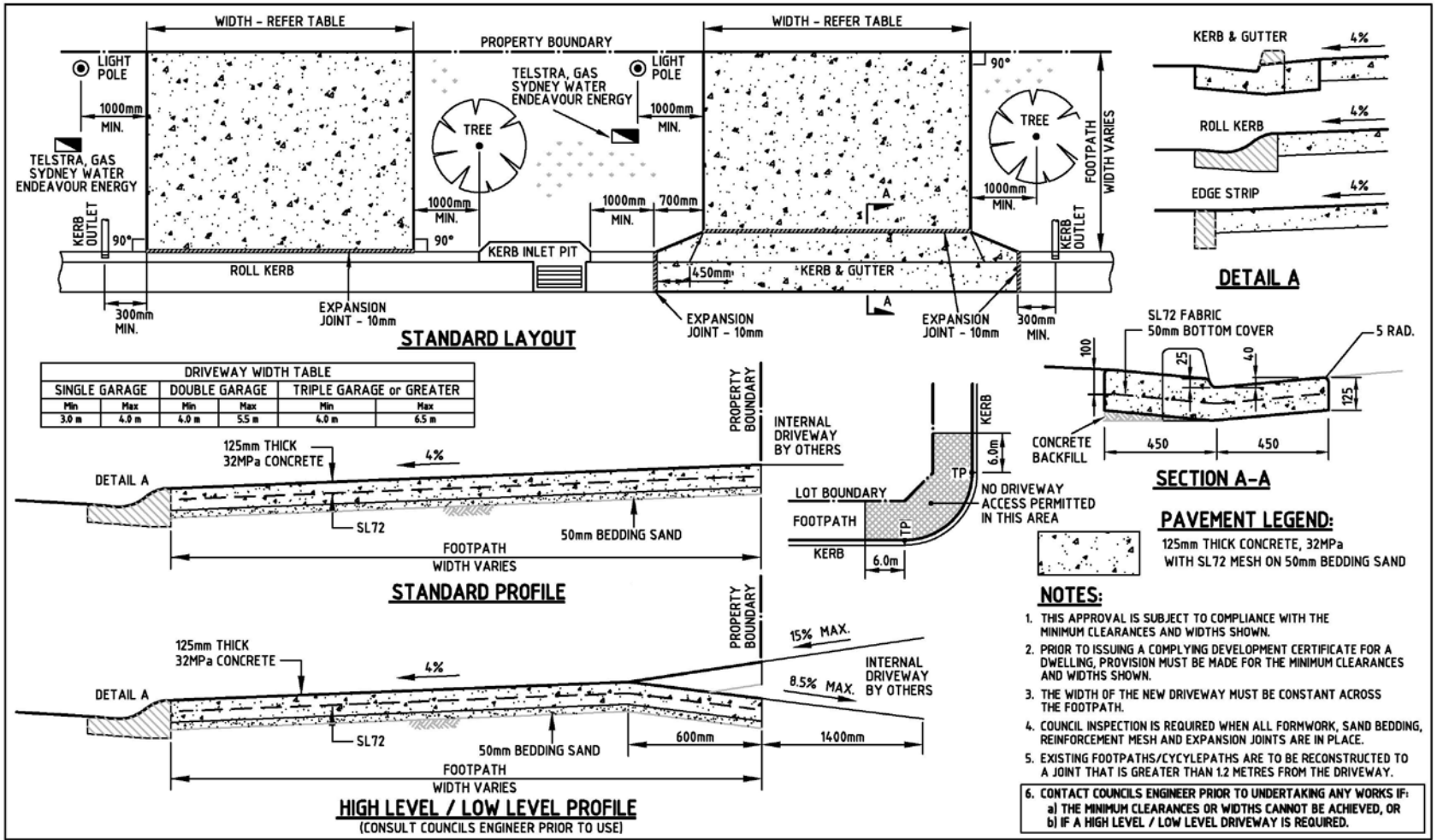
Bitumen Shoulder Seal

A bitumen shoulder seal is required from the concrete dish crossing to the edge of the existing bitumen road pavement. The pavement must consist of a compacted 150mm thick 2% lime stabilized DGB 20mm pavement with a bituminous 2 coat seal wearing course.

Useful Contact Telephone Numbers

Useful Contact Numbers

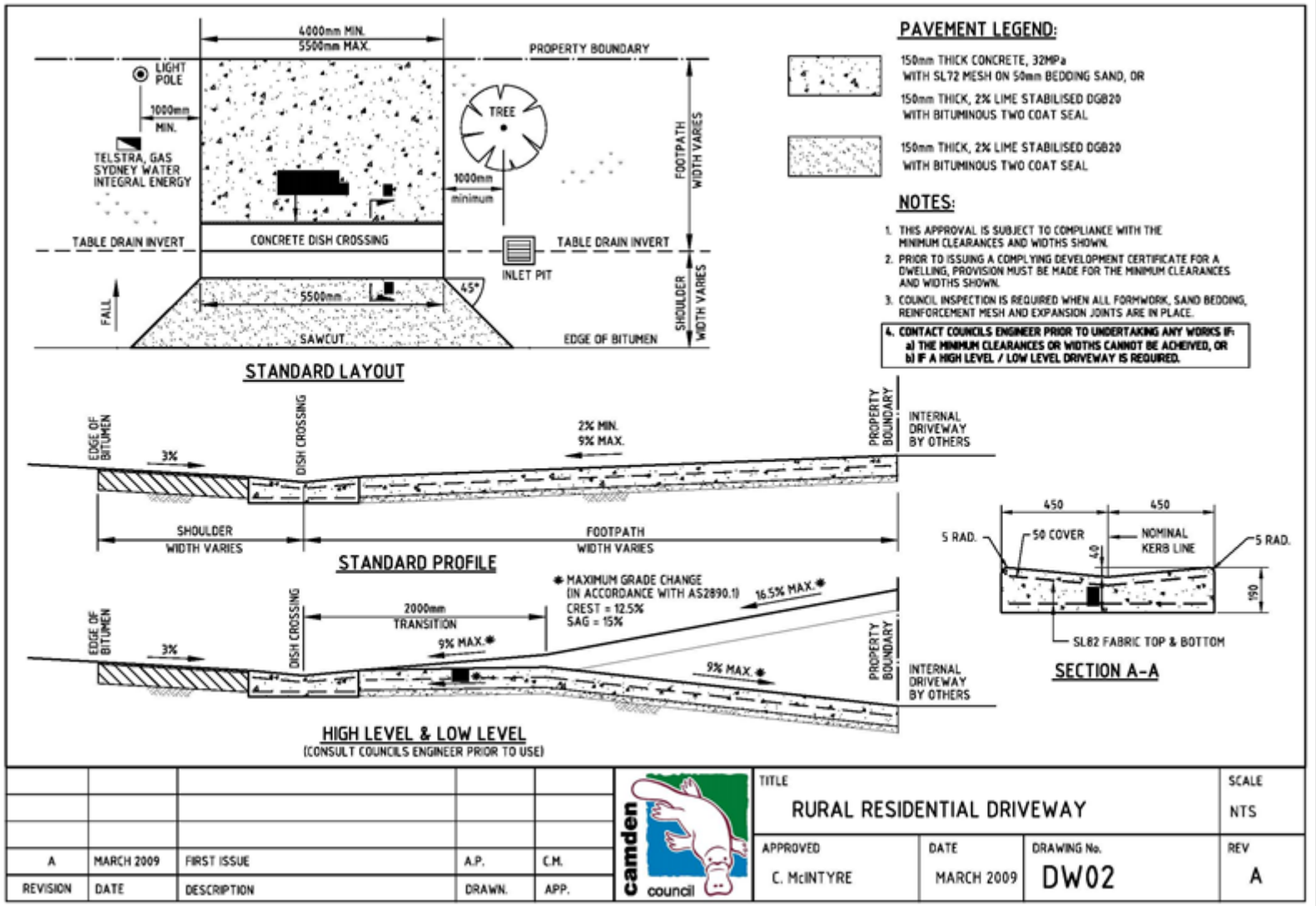
Camden Council	(02) 4654 7777
Endeavour Energy	131 081
Sydney Water	132 092
Telstra	132 203
Optus	1800 505 777
AGL	131 245
Dial Before You Dig (Cable Location)	1100



REVISION	DATE	DESCRIPTION	DRAWN.	APP.
D	OCT 2015	FOURTH ISSUE	G.C.	S.M.
C	MARCH 2011	THIRD ISSUE	A.P.	D.A.
B	MARCH 2010	SECOND ISSUE	A.P.	D.A.
A	MARCH 2009	FIRST ISSUE	A.P.	C.M.

camden council

TITLE			SCALE
URBAN RESIDENTIAL DRIVEWAY			NTS
APPROVED	DATE	DRAWING No.	REV
C. McINTYRE	MARCH 2009	**DW01**	D


4000mm MIN.
5500mm MAX.
PROPERTY BOUNDARY
LIGHT POLE
1000mm MIN.
TELSTRA, GAS
SYDNEY WATER
INTEGRAL ENERGY
TREE
1000mm minimum
FOOTPATH WIDTH VARIES
TABLE DRAIN INVERT
CONCRETE DISH CROSSING
TABLE DRAIN INVERT
INLET PIT
FALL
5500mm
45°
SHOULDER WIDTH VARIES
SAWCUT
EDGE OF BITUMEN
STANDARD LAYOUT
PAVEMENT LEGEND:
150mm THICK CONCRETE, 32MPa
WITH SL72 MESH ON 50mm BEDDING SAND, OR
150mm THICK, 2% LIME STABILISED DGB20
WITH BITUMINOUS TWO COAT SEAL
150mm THICK, 2% LIME STABILISED DGB20
WITH BITUMINOUS TWO COAT SEAL
NOTES:
1. THIS APPROVAL IS SUBJECT TO COMPLIANCE WITH THE MINIMUM CLEARANCES AND WIDTHS SHOWN.
2. PRIOR TO ISSUING A COMPLYING DEVELOPMENT CERTIFICATE FOR A DWELLING, PROVISION MUST BE MADE FOR THE MINIMUM CLEARANCES AND WIDTHS SHOWN.
3. COUNCIL INSPECTION IS REQUIRED WHEN ALL FORMWORK, SAND BEDDING, REINFORCEMENT MESH AND EXPANSION JOINTS ARE IN PLACE.
4. CONTACT COUNCILS ENGINEER PRIOR TO UNDERTAKING ANY WORKS IF:
a) THE MINIMUM CLEARANCES OR WIDTHS CANNOT BE ACHEIVED, OR
b) IF A HIGH LEVEL / LOW LEVEL DRIVEWAY IS REQUIRED.
EDGE OF BITUMEN
3%
DISH CROSSING
2% MIN.
5% MAX.
PROPERTY BOUNDARY
INTERNAL DRIVEWAY BY OTHERS
SHOULDER WIDTH VARIES
FOOTPATH WIDTH VARIES
STANDARD PROFILE
MAXIMUM GRADE CHANGE (IN ACCORDANCE WITH AS2890.1)
CREST = 12.5%
SAG = 15%
16.5% MAX.
2000mm TRANSITION
9% MAX.
9% MAX.
450
450
5 RAD.
50 COVER
NOMINAL KERB LINE
5 RAD.
40
150
SL82 FABRIC TOP & BOTTOM
SECTION A-A
HIGH LEVEL & LOW LEVEL
(CONSULT COUNCILS ENGINEER PRIOR TO USE)
REVISION
DATE
DESCRIPTION
DRAWN.
APP.
A
MARCH 2009
FIRST ISSUE
A.P.
C.M.
camden council
TITLE
RURAL RESIDENTIAL DRIVEWAY
SCALE
NTS
APPROVED
C. McINTYRE
DATE
MARCH 2009
DRAWING No.
DW02
REV
A